

Technical Document #162: Database Systems - Comprehensive Overview

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Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- NoSQL databases handle unstructured and semi-structured data.
- Database migration tools manage schema changes and data transformations.
- Database monitoring tracks performance metrics and identifies bottlenecks.